

CS391 Web Application Development

Project (2)

Assignment Date : 29.04.2026 09.00 AM

Deadline : 27.05.2026 11.59 PM (EXTENSION IS NOT POSSIBLE)

- This assessment must be prepared and finalized individually or a group.
- You are expected to develop a web application with ReactJS, NextJS, NodeJS and Bootstrap.
- When your project is completed, you must upload it to the LMS as an archive file. Don't add ***node_modules*** directory in the archive file.

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1. Project Details

Your second project will be very similar to your first project. One new page has been added, and the requirements have been revised to emphasize proper use of ReactJS components. Please read the instructions below carefully.

The project will contain two main pages: the **Home page** and the **Reservations** page. Other components will be associated with either the Home page or the Reservations page. Some components will be used as nested components.

You may change the location of the navigation bar (e.g., left sidebar or right sidebar).

Use React functional components for almost everything: HomePage, NavigationBar, SearchBar, HotelList, HotelCard, DealsAndDiscountsCarousel, DealAndDiscountCard, PopularSearches, PopularSearchCard, HotelDetail, Payment, ReservationsPage, SearchResultCarousel, SearchResults, etc.

Based on the user's selections, the Home page's component tree may look like the examples below. These are just suggestions to help you visualize the component structure. You can design different components.

Initial render of Home Page:

- <Home Page>
 - <SearchBar>
 - <DealsAndDiscountsCarousel>
 - <DealAndDiscountCard>
 - <PopularSearches>
 - <PopularSearchCard>

After searching

- <Home Page>
 - <SearchBar>
 - <SearchResultCarousel>
 - <HotelCard>
 - <PopularSearches>
 - <PopularSearchCard>

After searching for hotels, when the user clicks the "See More" link

- <Home Page>
 - <SearchBar>
 - <SearchResults>
 - <HotelCard>

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When user clicks over a hotel

- <Home Page>
 - <HotelDetail>

When user clicks the “book now” button

- <Home Page>
 - <HotelDetail>
 - <RoomSelection>

When user wants to complete the reservation

- <Home Page>
 - <PaymentInfo>

Before starting development, please **design your components on paper**. Decide on your **component tree**, each component’s **input parameters (props)**, **state variables**, and the **ownership of the data**.

Place the navigation bar in layout.js; as a result, the navigation bar will be included in all your pages.

1.1.Home Page (page.js)

The Home page is the starting point of your application. This page must be responsive and must **initially** contain the following sections: Search Bar, DealsAndDiscounts, SearchResultsCarousel, and Popular Searches. See **Figure 1** below for the screen design.

The page details are defined below:

1. **Search Bar:** This bar will be used for searching available hotels. It must contain four fields and all of them are required.
 - a. Search text field
 - b. Check-in and check-out date fields
 - c. Guest and room count selection fields
2. **Deals and Discounts Section:**
 - a. Retrieve deals list from the RESTful service (GET request).
 - b. Display at least **four deals or discounts**.
 - c. Generate the content of this section dynamically by iterating over your array.
 - d. The "Learn more" link does not need to redirect to another page.
3. **Popular Searches Section:**
 - a. Retrieve the **popular searches list** from the RESTful service (GET request).
 - b. Use the **Bootstrap carousel component** and display at least four items. You may show **one slide or multiple slides** on the screen at a time.
 - c. When a user clicks on a search item, display the **SearchResults** component with results for the selected search item. For example, if a user clicks on “Antalya,” the **SearchResults** component must show the hotels in Antalya. See Figure 3 below for the screen design.

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Home Page Usage Scenario:

- a. When a user clicks the Search button:
 - i. Retrieve the related hotels from the RESTful service (GET request).
 1. First, try to use the Query Capabilities provided by json-server (using query params, operators or advanced querying methods). If those are not sufficient, you may also use JavaScript's filter method. Use only the text entered in the search field. You cannot use dates, guest counts, or room data as search criteria, because your hotels array likely contains only hotel information, not reservation data.
 2. Your search function must check the user-entered search text in the **hotel title, description, or information fields**. Depending on your hotel data, you may also use other fields.
 - ii. Remove the content of the Deals and Discounts section and display the search results in a **Bootstrap Carousel** component. Display **five slides at a time** and sort the hotels by hotel name in ascending order.
 - iii. Each hotel card should contain an **image** and the **hotel name**. You may also display additional information about the hotel, such as **location, city, star count, rating count, and average rating**.
 - iv. Display a **"See more deals"** link at the top of the carousel.
 - v. See **Figure 2** below for the screen design
- b. When a user clicks **"See more deals"** link, remove Popular Searches and SearchResultCarousel sections and display all found hotels. Requirements are:
 - i. Hotels must be sorted by name in ascending order.
 - ii. See Figure 3 below for the screen design.
 - iii. The page must be responsive.
 - iv. Display at least four hotels per row.
 - v. When the user performs another search, do not display the initial Home page again; instead, rerender the component with the latest search results.
- b. When the user clicks on a hotel. Display the hotelDetails section. You can send the selected hotel ID to the HotelDetail component using props. In the component, you can retrieve the related hotel data from the RESTful service by sending a GET request. Requirements are:
 - i. The page must be responsive.
 - ii. See Figure 4 below for the screen design.
 - iii. Initially, the Room Selection section must be hidden.
 - iv. When the user clicks the Book Now button, display the Room Selection section. See Figure 5 below for the screen design.
 - v. You are free to modify the design of the Room Selection section.
 - vi. When the user clicks the Book Now button in the Room Selection section, display the Payment section. In this case, the room count and room type must be selected. You can send reservation data to the Payment component using props.
- c. Payment Section
 - i. Retrieve the selected hotel's reservation data from props.
 - ii. All form fields are required. Add any additional validation rules as needed. Validate the data before submitting the form and display error messages if there are any validation errors.

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- iii. You may optionally add additional fields for other guests' information, but this is not required.
- iv. When the user clicks the "Complete the Booking" button, create a reservation object and send object to the backend (send a POST request to your RESTful service).
- v. After the save completes successfully, clear all fields on the screen and show an alert box with a success message.
- vi. See Figure 6 below for the screen design
- vii. Your reservation object must contain:
 - 1. Hotel name and address
 - 2. Reservation data (room count, room type, dates)
 - 3. Guest data
 - 4. Payment information
- viii. A sample Reservation object structure is provided below:

```
Reservation: {
  "hotel": {
    "name": "Antalya Hotel",
    "address": "123 Beach Road, Antalya"
  },
  "reservationData": {
    "roomCount": 2,
    "guestCount": 4,
    "roomType": "Deluxe",
    "checkInDate": "2026-05-01",
    "checkOutDate": "2026-05-05"
  },
  "guestData": {
    "firstName": "John",
    "lastName": "Doe",
    "email": "john.doe@example.com",
    "phone": "+90 555 123 4567"
  },
  "paymentInformation": {
    "cardInfo": {
      "cardNumber": "1234-5678-9012-3456",
      "cardHolder": "John Doe",
      "expiryDate": "05/28",
      "cvv": "123"
    },
    "paymentMethod": "Credit Card",
    "totalAmount": 500
  }
}
```

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1.2. Reservations Page

When the user clicks the “Reservations” link in the navigation bar, display the “My Reservations” page. For retrieving reservations, send a GET request to your RESTful service and display the results.

When the user clicks the delete icon, remove the corresponding reservation from the list (send a DELETE request to your RESTful service). Your page must be rerendered.

See Figure 7 below for the screen design.

2. Data Requirements

For data management, use RESTful services. Do not use any external addresses for your data. You will create your own backend and a JSON file for your data. In this file, define four arrays:

- 1) Hotel list
- 2) Deals and discounts list
- 3) Popular searches list
- 4) Reservations list.

Include only the required fields in your JavaScript objects. Place your data file in the project and define a start script. You can use the same data you used in your first project. Keep the reservations list empty initially; you will add elements to this array when a reservation is completed.

Install the `npm-run-all` package in your project. Define an npm script using `npm-run-all` to run multiple npm scripts at the same time (for starting the application and the RESTful services). For more detailed information, you can refer to the presentations I have already uploaded over the next two weeks and the following links:

- <https://github.com/typicode/json-server>
- <https://github.com/mysticatea/npm-run-all>

3. Restrictions

- 3.1. Groups/individuals will be subjected to interviews to demonstrate mastery of the subject. Attending the demo is compulsory. If you do not attend, you will receive a 0 for the project, and as a result, you will FAIL.
- 3.2. Your application must be a ReactJS & NextJS application. Otherwise, you will **FAIL**.
- 3.3. Use only allowed technologies (e.g., do not use jQuery, React Router, Redux, etc.). If you are not sure, please ask me; otherwise, **you may lose points**.
- 3.4. For Bootstrap components, use ReactStrap or React-bootstrap packages.
- 3.5. Do not use local or session storage. For all data management operations, use the corresponding RESTful services (e.g., get hotels, save reservation, get deals, filter hotels, etc.). For sorting and filtering, first try to use the RESTful service endpoints provided by json-server. If those are not sufficient, you may also use JavaScript’s sort and filter methods.

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- 3.6. Always use the appropriate HTTP methods with the endpoint URL: use POST to save data, GET to retrieve data, PUT to update data, and DELETE to remove data.
- 3.7. Your application and RESTful services must start using a single script. Use the npm-run-all package to run scripts in parallel.
- 3.8. Do not update the DOM tree directly without using ReactJS/component logic.
- 3.9. Do not refer to images located in your local environment (e.g., C:\dev\images). You may use external image URLs or place images in your project (in this case, use relative URLs).
- 3.10. Do not upload the node_modules directory to the LMS or GitHub.
- 3.11. Before uploading your archive file to the LMS, unzip it in another directory, install dependencies, start, and test your application. If everything works correctly, you can upload your project.
- 3.12. Use React functional components for almost everything: HomePage, NavigationBar, SearchBar, HotelList, HotelCard, DealsAndDiscountsCarousel, DealAndDiscountCard, PopularSearches, PopularSearchCard, HotelDetail, Payment, ReservationsPage, SearchResultCarousel, SearchResults, etc.
- 3.13. Do not use class components; **otherwise, you will lose points.**
- 3.14. Use useState and useEffect hooks at least once.
- 3.15. Use Fetch or Axios to make HTTP requests to RESTful services.

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4. Screen Mockups

These mockup screens are provided for reference only. You may change the design as long as the required functions and specifications are met

4.1.Initial Home page

[Home](#)

My Reservations

Hotel
Antalya

Check-in/out
Jul 6 - Jul 11

Guests and rooms
2 Guests, 1 Room

Deals and Discounts



Earn 3K Bonus Points for every 3 nights

For a limited time, earn up to 28K Bonus Points across multiple stays. Register for Bonus Journeys and complete eligible nights between February 2 and April 15, 2026, at participating hotels. Terms apply.

[LEARN MORE →](#)



Earn Up To 50K Bonus Points

Stay longer, earn even more. Register now to get rewarded all year long with 5K Bonus Points on each qualifying stay of 5+ nights at Hyatt House and Hyatt Studios hotels worldwide. Complete stays by December 31, 2026. Terms apply.

[LEARN MORE →](#)



Turn your glamping getaway into 2K Bonus Points

World of Hyatt members can earn 2K Bonus Points per night on eligible stays at all 14 Under Canvas locations, as well as ULLUM Moab, The Fields of Michigan and Bar N Ranch. Register by June 30, 2026.

[LEARN MORE →](#)



Register to earn 3K Bonus Points per night

Relax and reconnect with family at Hyatt Vacation Club. Register by March 22, 2026, to earn 3,000 Bonus Points per night, up to a maximum of 30,000 Bonus Points, on qualifying stays completed between November 2, 2025, and March 29, 2026. Terms apply.

[LEARN MORE →](#)

Popular searches



Antalya

8,280 Hotels
£5,454 Avg.



Mugla Province

7,275 Hotels
£7,041 Avg.



Kappadokien

1,386 Hotels
£7,402 Avg.



Kuzey Kıbrıs Tür...

592 Hotels
£7,613 Avg.

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Figure 1: Home page: When user opens the home page first time

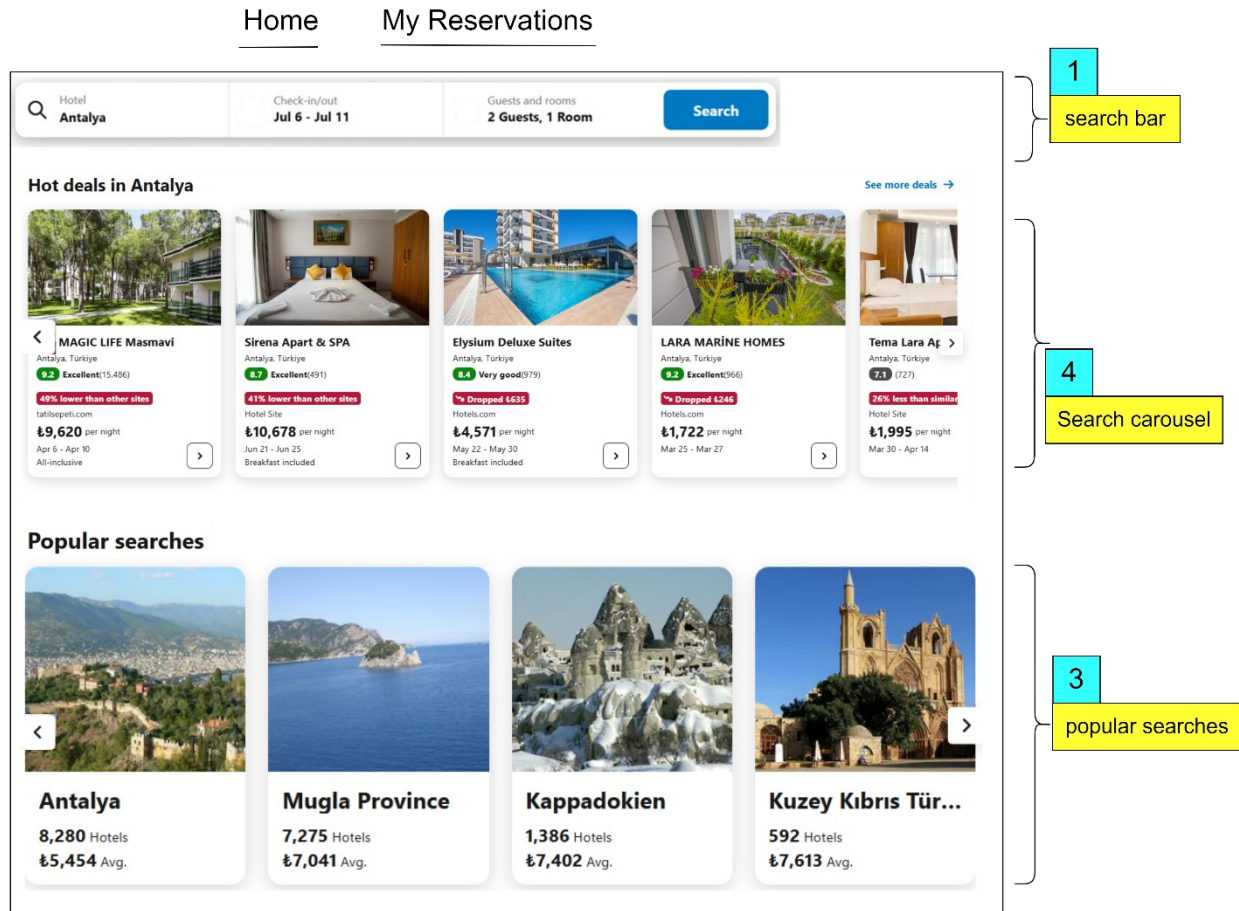


Figure 2: Home page: After search

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[Home](#) [My Reservations](#)

1	
search bar	

5	
Search results	

10

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4.3.Home Page - HotelDetails

Home

My Reservations

Suite Laguna

Muratpasa, Antalya - Show on Map

+14 photos

About the property

It is only 5 minutes by car to Konyaalti beach (Beach Park), which has the world's most famous, longest and cleanest beaches. Its distance to the sea is 400 meters by bird flight.
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Location

Our hotel has both city and holiday hotel facilities and features in the city center of Antalya. Our hotel; 15 Km to Antalya airport. at a distance, with its unique architectural structure reflecting the unique historical and cultural characteristics of the city of Antalya.

Hotel policies

Check/in

After 14:00

Check/out

Before 12:00

Pets

Pets not allowed

Smoking

No-smoking in the room

Child(ren)

Infants up to the age of 2 are free of charge.
1 child(ren) under the age of 3 are/is free of charge per room

6 Hotel photos carousel

7 Hotel rules

8 Hotel information

Figure 4: When user opens the HotelDetails Page first time

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[Home](#)

[My Reservations](#)

Suite Laguna

📍 Muratpasa, Antalya - [Show on Map](#)



+14 photos

About the property

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Hotel policies

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After 14:00

Check/out
Before 12:00

Pets
Pets not allowed

Smoking
No-smoking in the room

Child(ren)
Infants up to the age of 2 are free of charge.
1 child(ren) under the age of 3 are/is free of charge per room

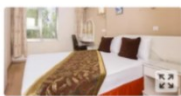
Room type	Conditions	Guests	2 per night	Select rooms	
 Standard Room 24m2 Air conditioning TV LCD TV Wireless Internet Free Early Check-in (After 09:00) - Late Check-out (Until 14:00)	Breakfast included Book Now, Pay Later FREE Cancellation Pay later with Zumbara	2	€ 8.891 All Taxes Included	0	5 Room 10 adults Total price € 78.236 All Taxes Included Book now € 3.912 2Money Book Now, Pay Later Book now, pay later.
 Junior Suite 35m2 Air conditioning Sitting group TV LCD TV Wireless Internet Free Early Check-in (After 09:00) - Late Check-out (Until 14:00)	Breakfast included Book Now, Pay Later FREE Cancellation Pay later with Zumbara	2	€ 11.113 All Taxes Included	2	
 Family Room 50m2 Air conditioning Sitting group	Breakfast included Book Now, Pay Later FREE Cancellation Pay later with Zumbara	2	€ 18.670 All Taxes Included	3	

Figure 5: HotelDetails Page when user clicks "Book now" button

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4.4.Home Page - Payment

Home

My Reservations

Your Reservation Information

Your Name *

First Name

Your Surname *

Last Name

Your Email Address *

Email

Your Phone Number *

+90 ✓ 555 555 55 55

☐ I approve [Clarification Text](#).

☐ I approve [Consent Form](#).

Payment Options

☰ Guarantee by credit card

Name on the card

👤 Name on the card

Card Number

☒ Card Number

Expiration Date

📅 MM/YY

Security Number

☒ CVC

☐ I have read [Preliminary Information Form](#) and [Distance Sales Contract](#) and I agree.

Complete the booking

10

Guest data

11

Payment data

Figure 6: Payment Page

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4.5. My Reservations Page

Home

My Reservations

Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026	Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026	Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026
Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026	Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026	Hotel Image Hotel Name Guest Name Room Count : 2 Guest Count : 4 Room Type : Deluxe Check In Date : 01.05.2026 Check Out Date : 05.05.2026

Figure 7: My Reservations Page

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5. RUBRIC

TOTAL	100
using npm-run-all	5
design of json-server data file	5
Navigation bar	8
Home Page	
Search Bar ve search functionality	10
Deals and Discounts	7
Popular Searches	8
Search Results Carousel	9
Search Results List	10
Hotel Details	8
Hotel Details with Room Selection section	10
Payment	10
My Reservations	
Listing reservations	5
Deleting a reservation	5